SEMINAR 1: Mean-Field Optimal Control
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Cucker-Smale Model

Felipe Cucker and Steve Smale (2007)

CS model

$$\begin{cases} \dfrac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \le i \le N, \\ \dfrac{d}{dt}v^i = \dfrac{K}{N}\displaystyle\sum_{j=1}^{N} \hat{\psi}\left(|x^j - x^i|\right)\left(v^j - v^i\right), \end{cases} \tag{1}$$

- x_i: position of i-th particle,
- v_i: velocity of i-th particle,
- $\hat{\psi}$: communication function,
- K: coupling strength,
- N: number of particles.

Kinetic CS Model

For large numbers of particles, say $N \to +\infty$, PDE can describe the system.

Kinetic CS Model (formal by BBGKY)

$$\begin{cases} \partial_t f + v\partial_x f + \partial_v \left(\kappa_1 L(f) f\right) = 0, \\ L(f) = \displaystyle\int_{\mathbb{R}^{2d}} \psi(|x-y|)(v^* - v) f(v^*, y, t) dv^* dy. \end{cases} \tag{2}$$

$f(x, v, t)$ denotes the density of particles at position x, with velocity v, at time t.

Mean Field Limit

How to describe the limit process? More precisely, how to describe the connection

$$(x_i(t), v_i(t)) \xrightarrow{N \to +\infty} f(x, v, t).$$

- Put $(x_i(t), v_i(t))$ and $f(x, v, t)$ in the same framework (space).
- Define a proper metric d on the space so that the distance between $(x_i(t), v_i(t))$ and $f(x, v, t)$ can be measured.
- Prove the distance between $(x_i(t), v_i(t))$ and $f(x, v, t)$ tends to 0, when N tends to ∞.

Definition

(S. Ha, J. Liu 2009) For $T \in [0, T)$, let $\mu \in L^\infty\left([0, T); \mathcal{M}\left(\mathbb{R}^{2d}\right)\right)$ be a measure valued solution with initial $\mu_0 \in \mathcal{M}\left(\mathbb{R}^{2d}\right)$ iff μ satisfies

1. (μ_t, g) is continuous as a function of t, for $g \in C_0(\mathbb{R}^{2d})$.
2. μ satisfies the integral equation: $g \in C_0^1(\mathbb{R}^{2d} \times [0, T))$

$$(\mu_t, g(\cdot,\cdot,t)) - (\mu_0, g(\cdot,\cdot,0)) = \int_0^t (\mu_s, \partial_s g + v \cdot \nabla_x g + F \cdot \nabla_v g) ds,$$

where $F(x, v, \mu_s) := -\int_{\mathbb{R}^{2d}} \psi(|x-y|)(v - v_*)\mu_s(dy, dv_*)$.

Connect ODE and PDE

(S. Ha, J. Liu 2009)

- (ODE) The solution of CS model (x_i, v_i) can be viewed as a measure valued solution of the kinetic CS model as

$$\mu_t^N := \frac{1}{N} \sum_i \delta(x - x_i(t))\delta(v - v_i(t)).$$

- (PDE) The classical solution of kinetic CS model is a measure valued solution.

Wasserstein Distance (Metric)

Define $\mathcal{P}_1$ to be of measures μ such that for some $x_0 \in X$,

$$\int_X |x - x_0| d\mu(x) < \infty.$$

Then, the Wasserstein-1 distance between the two probability measures μ and ν in $\mathcal{P}_1(X)$ is defined by

Definition

$$W_1(\mu, \nu) := \inf_{\gamma \in \Gamma(\mu,\nu)} \int_{X \times X} |x - y| d\gamma(x, y), \tag{3}$$

where $\Gamma(\mu, \nu)$ denotes the collection of all measures in $X \times X$ with the marginals μ and ν.

Rigorous Mean Filed Limit

The rigorous proof follows the results below:

S. Ha, J. Liu 2009

- Suppose $\mu, \nu \in L^\infty([0, T); \mathcal{M}(\mathbb{R}^{2d}))$ are two measure value solution of kinetic CS model with initial datum (μ_0, ν_0). (μ_0, ν_0) are compact supported with finite moments up to 2. Then there exists a constant $C(T)$ such that

$$W_1(\mu_t, \nu_t) \leq C(T) W_1(\mu_0, \nu_0).$$

- For any $\mu_0 \in \mathcal{M}(\mathbb{R}^{2d})$, there exists a sequence μ_0^N such that

$$\lim_{N \to \infty} W_1(\mu_0, \mu_0^N) = 0.$$

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ODE

ODE Model

$$
\begin{cases}
\frac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \leq i \leq N, \\
\frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f(t, x_i, v_i), \quad i = 1, \cdots N, \quad t \in [0, T], \\
\mu_N = \frac{1}{N}\sum_{j=1}^{N} \delta_{(x_i, v_i)}.
\end{cases}
\tag{4}
$$

μ_N is the empirical measure supported on the agents $(x_i, v_i) \in \mathbb{R}^{2d}$, $f(t, x_i, v_i)$ is the a controller and is vector valued function i.e

$$f(t, x, v) : \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d.$$

Cost Function

For any controller $f(t, x, v)$ of the system, we can define the cost functional:

Cost Function

$$\mathcal{E}_{\psi}^{N}(f) := \int_0^T \int_{\mathbb{R}^{2d}} \left(L(x, v, \mu_N(t)) + \psi(f(t, x, v))\right) d\mu_N(t)(x, v)dt,$$

where we require

- $\psi : \mathbb{R}^d \to [0, +\infty]$ is a nonnegative convex function.
- $H : \mathbb{R}^{2d} \to \mathbb{R}^d$ is a sublinear and locally Lipschitz continuous interaction kernel.
- $L : \mathbb{R}^{2d} \times \mathcal{P}_1(\mathbb{R}^{2d}) \to \mathbb{R}_+$ is a continuous function with respect to the product topology.

Cucker-Smale (example)

- $\psi(\cdot) = \gamma|\cdot|$.
- $H(x, v) = a(|x|)v$, where a is a non-increasing smooth positive function.
- $L(x, v, \mu_N) = |v - (\int_{\mathbb{R}^{2d}} w d\mu_N(y, w))|^2 = |v - \frac{1}{N}\sum_{j=1}^{N} v_j|^2$.

CS Model with Controller

$$\begin{cases} \dfrac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \leq i \leq N, \\ \dfrac{d}{dt}v^i = \dfrac{K}{N}\displaystyle\sum_{j=1}^{N} a\left(|x^j - x^i|\right)\left(v^j - v^i\right) + f(t, x_i, v_i). \end{cases} \tag{5}$$

Cost Function

$$\mathcal{E}_{\psi}^{N}(f) = \int_0^T \frac{1}{N}\sum_{i=1}^{N}\left(|v - \frac{1}{N}\sum_{j=1}^{N} v_j|^2 + \gamma|f(t, x_i, v_i)|\right) dt.$$

Optimal Control Problem

Let $F([0,T])$ be the collection of all controller $f(t,x,v)$, if there exists a control f_∞ such that $(x_i^\infty(t), v_i^\infty(t))$ solves the equation

Solution Exists

$$\begin{cases} \frac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \le i \le N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f_\infty(t, x_i, v_i), \quad i = 1, \cdots N, \quad t \in [0,T]. \end{cases} \tag{6}$$

Minimum Cost

$$\mathcal{E}_\psi^N(f_\infty) = \min_{f \in F([0,T])} \mathcal{E}_\psi^N(f).$$

PDE Model

When N tends to infinity, like the non-control problem, we have the PDE model

PDE

$$\mu_t + v \cdot \nabla_x \mu = \nabla_v \cdot [(H * \mu + f)\mu].$$

and the cost function defined as

Cost Function

$$\mathcal{E}_\psi(f) := \int_0^T \int_{\mathbb{R}^{2d}} (L(x, v, \mu(t)) + \psi(f(t, x, v)))\, d\mu(t)(x, v) dt,$$

Optimal Control problem

Let $F([0, T])$ be the collection of all controller $f(t, x, v)$, if there exists a control f_∞ such that μ_∞ is a measure valued solution of the equation

PDE

$$\mu_t + v \cdot \nabla_x \mu = \nabla_v \cdot [(H * \mu + f)\mu].$$

Moreover

Minimum Cost

$$\mathcal{E}_\psi(f_\infty) = \min_{f \in F([0,T])} \mathcal{E}_\psi(f).$$

Hint

To describe the mean-field limit we need to clarify

- The existence of the control problem in both ODE and PDE level.
- The existence of the optimal control f_∞ for both ODE and PDE model.
- Connect ODE and PDE solutions in the sense of measure valued solution and show the mean-field limit.

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Admissible Controls

Definition

For a horizon time $T > 0$, and an exponent $1 \leq q < +\infty$, we fix a control bound function $\ell \in L^q(0, T)$. The class off admissible control functions $\mathcal{F}_\ell([0, T])$ is so defined: $f \in \mathcal{F}_\ell([0, T])$ if and only if

- $f : [0, T] \times \mathbb{R}^n \to \mathbb{R}^d$ is a Caratheodory function,
- $f(t, \cdot) \in W^{1,\infty}_{loc}(\mathbb{R}^{2d}, \mathbb{R}^d)$ for almost every $t \in [0, T]$, and
- $|f(t, 0)| + Lip(f(t, \cdot), \mathbb{R}^d) \leq \ell(t)$ for almost every $t \in [0, T]$.

Where Caratheodory function means

Caratheodory Function

- For fixed x, $f(\cdot, x)$ is measurable in Ω_x, the cross section of x.
- For fixed t, $f(t, \cdot)$ is continuous in Ω_t, the cross section of t.

Caratheodory Differential Equation and Solution

Consider the differential equation

$$\dot{y}(t) = g(t, y(t)), \quad t \in [0, T], \quad y \in \Omega \subseteq \mathbb{R}^n.$$

Definition

We call the equation is Caratheodory if $g(t, y)$ is a Caratheodory function.

Definition

For $0 < \tau \leq T$, we call $y : [0, \tau] \to \Omega$ a Caratheodory solution to the equation on $[0, \tau]$ if

- y is absolutely continuous,
- y satisfies the equation a.e in $[0, \tau]$.

Local Existence

Theorem

(Filippov A.1)

- *Consider Caratheodory equation, if there exists a function $m \in L^1((0,T))$ such that*

$$|g(t,y)| \leq m(t)$$

  for a.e $t \in [0,T]$ and $y \in \Omega$. Then given $y_0 \in \Omega$, there exists $0 < \tau \leq T$ and a Caratheodory solution $y(t)$ on $[0,\tau]$ with initial data $y(0) = y_0$.

- *More over, if there exists $\ell \in L^1((0,T))$ such that*

$$|g(t,y_1) - g(t,y_2)| \leq \ell(t)|y_1 - y_2|.$$

  Then the Caratheodory solution is unique on $[0,\tau]$.

Theorem

(A.2) Consider an interval $[0, T]$. Assume there exists a function $m \in L^1((0, T))$ such that

$$|g(t, y)| \leq m(t)(1 + |y|)$$

for a.e. $t \in [0, T]$ and $y \in \mathbb{R}^n$. Then for any $y_0 \in \mathbb{R}^n$, there exists a solution $y(t)$ defined on $[0, T]$ with $y(0) = y_0$. The solution satisfies

$$|y(t)| \leq \left(|y_0| + \int_0^t m(s)ds \right) e^{\int_0^t m(s)ds}.$$

Proof:

Verification

Consider our equation

$$\begin{cases} \dfrac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \le i \le N, \\ \dfrac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f(t, x_i, v_i), \quad i = 1, \cdots N, \quad t \in [0, T], \\ \mu_N = \dfrac{1}{N}\displaystyle\sum_{j=1}^{N} \delta_{(x_i, v_i)}. \end{cases} \tag{7}$$

It is easy to check the right hand side satisfies previous theorem.
Proof:

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Outline of Construction

Outline

- Find a control f_∞.
- Prove the corresponding control problem admits a Caratheodory solution.
- Prove f_∞ is a minimizer

$$\mathcal{E}_\psi^N(f_\infty) = \min_{f \in F([0,T])} \mathcal{E}_\psi^N(f).$$

Idea of Construction of f_∞

As the cost functional is non-negative, for any control $f(t,x,v)$, there is a natural lower bound zero for $\mathcal{E}_\psi^N(f)$. Thus there exists a minimizing sequence $\mathcal{E}_\psi^N(f_k)$ such that

$$\lim_{k\to\infty} \mathcal{E}_\psi^N(f_k) = \inf \mathcal{E}_\psi^N.$$

In order to apply compactness to obtain the optimal control, we need the structure of the space of controllers i.e $\mathcal{F}_\ell([0,T])$.

Remark 2.2

Suppose $f \in \mathcal{F}_\ell([0,T])$, then we have $f(t)$ is measurable for any fixed t and

$$f \in L^q((0,T), W^{1,p}(\Omega, \mathbb{R}^d))$$

for every open bounded subset $\Omega \subset \mathbb{R}^n$, $1 < p < +\infty$ and q is the integrability exponent of ℓ.

Proof:

Description of $\mathcal{F}_\ell([0,T])$

Remark 2.3

Consider the space

$$C_{\ell,\Omega} := \{f \in L^q((0,T), W^{1,p}(\Omega,\mathbb{R}^d)) : |f(t,0)| + Lip(f(t,\cdot),\mathbb{R}^d) \le \ell(t)\},$$

a.e. $t \in [0,T]$. If $f \in C_{\ell,\Omega}$ for all $\Omega \subset \mathbb{R}^n$, then $f \in \mathcal{F}_\ell$.

We only need to show f is Caratheodory. $f(t,\cdot)$ continuous because of the property of $C_{\ell,\Omega}$. Apply the equivalence of weak measurable and measurable as in Remark 2.2 to have $\langle f(t), x_0\rangle = f(x_0,t)$ is measurable.

Find f_∞

Theorem 2.5

Let X be a reflexive and separable Banach space. Let (f_j) be a sequence of functions in $L^q((0,T),X)$. Suppose there exists $m \in L^q([0,T])$ such that $||f_j(t)||_X \leq m(t)$ for a.e $t \in [0,T]$. Then there exists a subsequence (f_{jk}) and a function $f \in L^q([0,T])$ such that

$$\lim_{k\to\infty} \int_0^T \langle \phi(t), f_{jk}(t,\cdot) - f(t,\cdot)\rangle = 0,$$

for all $\phi \in L^{q'}((0,T),X')$ and

$$w - \lim_{k\to\infty} \int_{t_1}^{t_2} f_{jk}(t)dt = \int_{t_1}^{t_2} f(t)dt, \quad t_1, t_2 \in [0,T].$$

The integrals are in the sense of Bochner.

Proof:

Find f_∞

Theorem 2.6

Let Ω be a bounded smooth and open subset of $\mathbb{R}^n$. Assume (f_j) be a sequence of functions in $L^q((0,T), W^{1,p}(\Omega, \mathbb{R}^d))$ such that

$$|f_j(t,0)| + Lip(f_j(t,\cdot),\Omega) \leq \ell(t) \quad a.d.t \in [0,T].$$

Then there exists a subsequence (f_{jk}) and a function $f \in L^q((0,T), W^{1,p}(\Omega, \mathbb{R}^d))$ such that

$$w - \lim_{k\to\infty} f_{jk} = f.$$

weakly in $L^q((0,T), W^{1,p}(\Omega, \mathbb{R}^d))$ and

$$|f(t,0)| + Lip(f(t,\cdot),\Omega) \leq \ell(t) \quad a.d.t \in [0,T].$$

Find f_∞

Corollary 2.7

Assume (f_j) be a sequence of functions in $\mathcal{F}_\ell$ for a given function ℓ. Then there exists a subsequence and function $f \in \mathcal{F}_\ell$ such that

$$\lim_{k\to\infty} \int_0^T \langle \phi(t), f_{jk}(t,\cdot) - f(t,\cdot)\rangle dt = 0,$$

for all $\phi \in L^{q'}((0,T), H^{-1,p}(\mathbb{R}^n, \mathbb{R}^d))$ such that $supp(\phi) \subset \Omega$ for all $t \in [0,T]$, where Ω is relative compact.

Now consider the minimizing sequence, then the weak limit is our construction of f_∞.

Find (x_∞, v_∞)

Lemma 3.2

Let $f \in \mathcal{F}_\ell$ and (x, v) be the solution of control problem with initial data $(x(0), v(0))$. Then

$$\mathcal{V}(t) \leq \{\mathcal{V}(0) + (1 + \mathcal{X}(0)) \left(2CT + \int_0^t \ell(s)ds\right) \} e^{(1+T)\int_0^t (2C+\ell(s))ds}.$$

Where

$$\mathcal{V}(t) = \max_N |v_i|, \quad \mathcal{X}(t) = \max_N |x_i|.$$

Then the trajectory $(x(t), v(t))$ is uniformly bounded.

Find (x_∞, v_∞)

With the equi-boundedness and equi-absolute continuity of $v_i^k(t)$, we can imply equi-Lipschitz of $x_i^k(t)$ and equi-integrability of $\dot{v}_i^k(t)$. Then by Ascoli-Arzela theorem there exists a subsequence and an absolutely continuous trajectory $(x(t), v(t))$ such that

- $x_i^k \to x_i, \quad t \in [0, T],$
- $v_i^k \to v_i, \quad t \in [0, T],$
- $\dot{x}_i^k \to \dot{x}_i, \quad t \in [0, T],$
- $\dot{v}_i^k \rightharpoonup \dot{v}_i, \quad L^q([0, T]).$

Now we are going to show the constructed (x_i, v_i) is a Caratheodory solution. As we already show (x_i, v_i) are absolute continuous, we only need to show that

ODE

$$\begin{cases} \frac{d}{dt}x^i = v^i, \quad t > 0, \quad 1 \le i \le N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f_\infty(t, x_i, v_i), \quad i = 1, \cdots N, \quad t \in [0, T], \\ \mu_N = \frac{1}{N}\sum_{j=1}^{N} \delta_{(x_i, v_i)}. \end{cases} \tag{8}$$

Caratheodory Solution

We have four terms to deal with

Convergence for Different Terms

- For the first equation, we apply uniformly convergence and can obtain the resluts.
- Lemma A.7 shows $H * \mu_N^k(x_i^k, v_i^k) \to H * \mu_N(x_i, v_i)$.
- $\dot{v}_i^k \rightharpoonup \dot{v}_i, \quad L^q([0, T])$.
- The control term $f(t, x, v)$ is the difficult term.

For the second equation it is equivalent to consider

$$\xi \cdot \int_0^{\bar{t}} \dot{v}_i(t)dt = \xi \cdot \int_0^{\bar{t}} [(H * \mu_N) + f(t, x_i(t), v_i(t))]dt.$$

From previous analysis, we only need to prove

$$\lim \xi \cdot \int_0^{\bar{t}} f_k(t, x_i^k(t), v_i^k(t))dt = \xi \cdot \int_0^{\bar{t}} f(t, x_i(t), v_i(t))dt.$$

Proof:

Optimal Control

Now we want to prove the constructed f_∞ is an optimal control i.e we want to prove f_∞ is a minimizer

optimal

$$\mathcal{E}_\psi^N(f_\infty) = \min_{f \in F([0,T])} \mathcal{E}_\psi^N(f).$$

Idea: from the construction, $f = \omega - \lim_{k\to\infty} f_{n_k}$, where $\{f_{n_k}\}$ is a subsequence of minimizing sequence $\{f_n\}$. Thus we only need to prove

optimal attained

$$\mathcal{E}_\psi^N(f_\infty) \leq \liminf_{k\to\infty} \mathcal{E}_\psi^N(f_{n_k}).$$

Running cost

Now we consider

$$\mathcal{E}_\psi(f) := \int_0^T \int_{\mathbb{R}^{2d}} \left(L(x, v, \mu(t)) + \psi(f(t, x, v)) \right) d\mu(t)(x, v) dt,$$

the running cost is $L(x, v, \mu(t))$. By assumption of the continuous of L and the uniformly convergence of $(x_i^k(t), v_i^k(t), \mu_N^k(t))$ to $(x_i(t), v_i(t), \mu_N(t))$, we have

running cost

$$\lim_{k\to\infty} \int_0^T \int_{\mathbb{R}^{2d}} L(x, v, \mu_N^k(t)) d\mu_N^k(t) dt = \int_0^T \int_{\mathbb{R}^{2d}} L(x, v, \mu_N(t)) d\mu_N(t) dt.$$

Proof: By continuity of L.

Next for the term

$$\int_0^T \int_{\mathbb{R}^{2d}} \psi(f(t,x,v))d\mu(t)(x,v)dt,$$

we have

penalization

$$\liminf_{k\to\infty} \int_0^T \int_{\mathbb{R}^{2d}} \psi(f_k(t,x,v))d\mu_N^k(t)dt \geq \int_0^T \int_{\mathbb{R}^{2d}} \psi(f(t,x,v))d\mu_N(t)dt.$$

Proof: By Theorem 2.12.

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Outline

- What is measure valued solution? (Definition)
- How to construct the measure valued solution? (Limit)
- Is the limit actually a measure valued solution? (Existence)
- Is the solution optimize the cost function? (Optimization)

Definition

Definition

Let $\ell \in L^1(0,T)$. Fix a function f belonging to the class $\mathcal{F}_\ell$. Given a Locally Lipschitz function H, we say that $\mu : [0,T] \to \mathcal{P}_1(\mathbb{R}^{2d})$ continuous with $\mathcal{W}_1$ is a weak equi-compactly supported solution of the equation if there exists $R > 0$ such that

$$supp(\mu(t)) \subset B(0,R),$$

for every $t \in [0,T]$, and, defining $w_{H,\mu,f}(t,x,v) = (v, H * \mu + f)$, one has

$$\frac{d}{dt}\int_{\mathbb{R}^{2d}} \xi(x,v)d\mu(t) = \int_{\mathbb{R}^{2d}} \nabla\xi(x,v)\cdot w_{H,\mu,f}(t,x,v)d\mu(t)$$

for every $\xi \in C_c^\infty(\mathbb{R}^{2d})$, in the sense of distribution.

Construction

lemma 4.3 (equi-compact support solution of ODE)

Consider the empirical measure as the measure valued solution of ODE

$$\mu_N(t)(x,v) = \frac{1}{N}\sum_{i=1}^{N} \delta_{(x_i,v_i)}(x,v),$$

with $supp(\mu_N(0)) \subset B(0,R_0)$. Then we have

- $supp(\mu_N(t)) \subset B(0,R_T)$, where R_T is independent of N.
- $\mathcal{W}_1(\mu_N(t_1),\mu_N(t_2)) \le L_T \int_{t_1}^{t_2}(1+\ell(s))ds$ for all $t_1,t_2 \in [0,T]$.
- For all $\hat{t} \in [0,T]$ and for all $\xi \in C_c^1(\mathbb{R}^{2d})$

$$\langle \xi, \mu_N(\hat{t}) - \mu_N(0)\rangle = \int_0^{\hat{t}}\int_{\mathbb{R}^{2d}} \nabla\xi(x,v)\cdot w_{H,\mu_N,f_N}(t,x,v)d\mu_N(t)dt$$

Proof:

Construction of solution

Construction of solution

Suppose μ_N^0 are equi-bounded and $\mathcal{W}_1(\mu_N^0, \mu^0) \to 0$ and the corresponding solutions are denoted by $\mu_N(t)$ with control f_N.
Then there exist a subsequence f_{N_k} converge in the sense of (2.7) to f; the corresponding μ_{N_k} converge in Wasserstein distance to μ which is equi-compact with μ_{N_k} and solve the equation with control f and initial data μ_0. Moreover, we have the inequality

$$\mathcal{E}_\psi(f) \leq \liminf_{k\to\infty} \mathcal{E}_\psi^{N_k}(f_{N_k}).$$

Proof: Thm 4.4 and Cor 4.6.

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Optimal control

Theorem 4.7

Suppose μ_N^0 are equi-bounded and $\mathcal{W}_1(\mu_N^0, \mu^0) \to 0$ and the corresponding solutions are denoted by $\mu_N(t)$ with common control f.
Then μ_N converge in Wasserstein distance to μ which is equi-compact with μ_N and solve the equation with control f and initial data μ_0.
Moreover, we have the equality

$$\mathcal{E}_\psi(f) = \lim_{N\to\infty} \mathcal{E}_\psi^N(f).$$

Proof: Same as Thm 4.4. Thm A.3, Thm A.5, Thm A.6, Thm A.7 imply Thm A.8. Then we conclude not only subsequence convergence but the whole sequence convergence.

optimal control

Suppose μ_N^0 are equi-bounded and $\mathcal{W}_1(\mu_N^0, \mu^0) \to 0$ and the corresponding solutions are denoted by $\mu_N(t)$ with control f_N.
Then there exist a subsequence f_{N_k} converge in the sense of (2.7) to f; the corresponding μ_{N_k} converge in Wasserstein distance to μ which is equi-compact with μ_{N_k} and solve the equation with control f and initial data μ_0. Moreover, we have

$$\mathcal{E}_\psi(f) \leq \mathcal{E}_\psi(g), \quad g \in \mathcal{F}_\ell.$$